Chemical Histories of Dark Clouds: Establishing the Survival Timescale of the Eos Molecular Cloud

Abstract

CO-dark molecular clouds, H_2-dense regions that cannot be detected in CO emission, comprise a significant fraction (~40%) of the Milky Way's molecular mass. As such, the chemical history of CO-dark clouds is crucial to our understanding of the evolution of the Milky Way's interstellar medium. However, the survival timescales of CO-dark clouds are unknown: velocity structures within the clouds dissipate rapidly, leaving quiescent gas consistent with both transient and long-lived structures. We propose high-resolution JWST/NIRSpec observations of the newly discovered CO-dark cloud Eos (~94 pc away in the Local Bubble) to determine whether its gas is chemically young, out of equilibrium, or fully relaxed. JWST is the only facility capable of detecting faint H_2 rotational emission, enabling use of the H_2 ortho-to-para ratio (OPR) as a chemical clock to trace thermal processing in Eos' quiescent H_2 gas. These observations will directly test intermittent turbulence models and establish Eos as a benchmark for the survival timescale of CO-dark material in the Milky Way. Significantly, this program will establish whether CO-dark clouds have chemical histories consistent with CO-bright molecular clouds, clarifying the role of CO-dark gas in shaping the initial conditions for star formation across the galaxy.

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- **Scientific Justification**

Establishing the Chemical History of an Invisible ISM Reservoir

CO-dark molecular clouds contain a significant fraction of the Milky Way's molecular mass (~40%; Smith et al. 2014), yet their chemical histories remain largely unconstrained. A large fraction of interstellar H_2 is invisible to CO-based surveys; its existence is instead typically inferred from [CII] line emission and gamma-rays generated by cosmic-ray interactions with diffuse gas (Pineda et al. 2013; Grenier et al. 2005). The CO-dark phase arises where H_2 self-shields against photodissociation, but CO is destroyed, producing extended molecular envelopes undetectable by traditional tracers (Wolfire et al. 2010). Nearby examples within the Local Bubble confirm that these clouds harbor significant molecular mass reservoirs that are systematically missed by CO surveys (Langer et al. 2014). <u>Because CO-dark gas accounts for nearly half of the molecular ISM, its chemical state fundamentally shapes our understanding of ISM evolution and the initial conditions for star formation across the Galaxy.</u>

However, the survival timescale and chemical evolution of CO-dark clouds remain poorly constrained: standard kinematic and density diagnostics alone cannot distinguish between turbulence-driven structures and long-lived, chemically relaxed gas (Mac Low & Klessen 2004; Glover & Mac Low 2007). Turbulent velocity structures dissipate on timescales far shorter than those over which chemical networks equilibrate, leaving no kinematic record of prior thermal or chemical processing. Fortunately, this problem is now tractable. The discovery of Eos provides a nearby laboratory, turbulence models now offer testable predictions, and JWST uniquely enables the required chemical diagnostic.

<u>We propose high-resolution JWST/NIRSpec spectroscopy of H_2 rotational emission in the Eos molecular cloud to measure the H_2 ortho-to-para ratio (OPR)</u>, a uniquely long-lived chemical clock capable of distinguishing chemically young, non-equilibrium, and fully relaxed gas. No prior or current facility can execute this measurement: for example, Spitzer and SOFIA lacked the sensitivity for faint, low-excitation H_2 lines in cold quiescent gas, and HST carries no spectroscopic capability at the relevant near-infrared wavelengths. We will observe two regions within Eos—one dense and well-shielded, one diffuse—to sample OPR across environments and classify Eos into one of three chemical regimes. **This program will deliver the first direct constraint on the chemical age of CO-dark molecular gas**, breaking a long-standing observational degeneracy and placing empirical bounds on CO-dark cloud survival timescales. Publicly released JWST/NIRSpec spectra and derived OPR measurements will serve as a community benchmark, anchoring future ISM programs and informing models of molecular cloud assembly, star formation efficiency, and baryon cycling in galaxies.

The Chemical Age of CO-dark Clouds Has Not Been Determined

CO-dark molecular clouds are identified through indirect tracers ([CII] emission, gamma-ray observations, dust extinction, kinematics) that constrain their current mass, extent, and dynamical history. However, these well-established diagnostics carry no record of chemical age: they are equally consistent with recently processed, chemically young material or long-lived, fully relaxed gas. This degeneracy arises because turbulent velocity

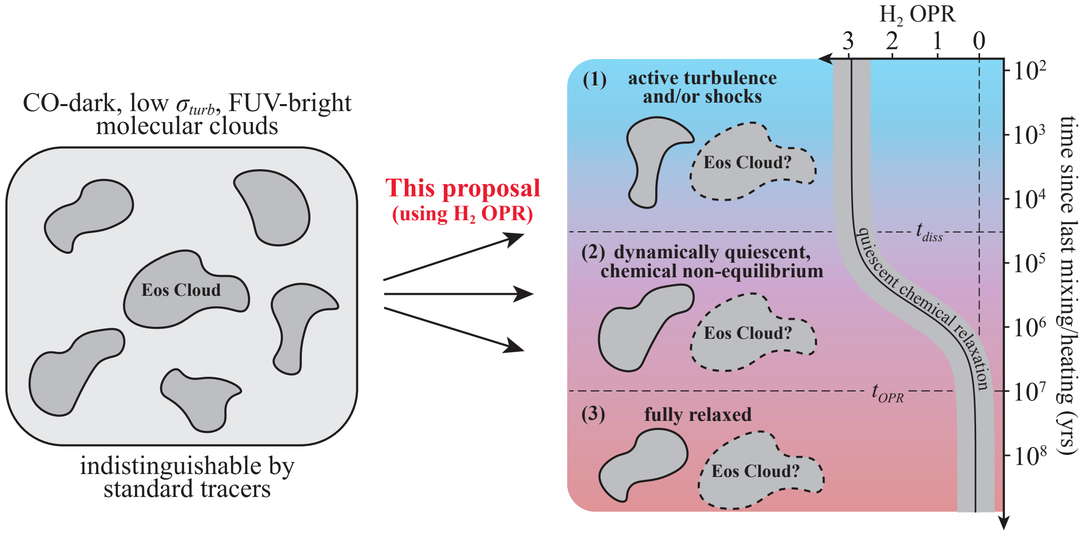

Figure 1: **The H_2 ortho-to-para ratio (OPR) provides a uniquely long-lived chemical record that can reveal the hidden dynamical history of abundant CO-dark clouds such as the newly discovered Eos cloud** (Flower et al. 2006). Left: CO-dark molecular clouds that are dynamically indistinguishable using standard tracers. Right: Evolution of H_2 OPR as a function of time since the last significant mixing or heating event. The solid curve shows quiescent chemical relaxation from the value at formation toward thermal equilibrium. Horizontal dashed lines denote the kinematic dissipation timescale (t_{diss}) and the much longer OPR equilibration timescale (t_{OPR}). H_2 OPR measurements can classify dark gas into three different regimes (Glover & Mac Low 2007): (1) chemically young, (2) dynamically quiescent yet chemically non-equilibrium, and (3) chemically old. Dark clouds such as Eos occupy an observationally degenerate state where velocity information alone cannot distinguish between recently processed gas and long-lived remnants. This program will measure H_2 OPR across Eos to determine the chemical age of its constituent gas. Doing so will enable us to constrain the survival timescale of dark molecular gas and directly evaluate turbulence models for the diffuse ISM.

structures dissipate on timescales of ~10^4 yr, far shorter than the timescales over which chemical networks equilibrate (Mac Low & Klessen 2004; Glover & Mac Low 2007; Liszt 2007). **The chemical histories of CO-dark clouds thus remain a fundamental open question.** Given that dark gas accounts for nearly half of the molecular ISM, whether these clouds are transient turbulent byproducts or long-lived molecular reservoirs drives our understanding of molecular cloud assembly and the initial conditions for star formation.

The H_2 ortho-to-para ratio (OPR) is the key chemical diagnostic needed to answer these questions. Set during H_2 formation and evolving towards its chemical equilibrium value via slow proton-exchange reactions, OPR evolves on timescales of ~10^7 yrs and preserves a record of a cloud's chemical history long after its kinematic and density structure have relaxed (Flower et al. 2006; Pagani et al. 2011). In warm or recently shocked gas, the OPR approaches its high-temperature statistical limit of 3; in cold, long-lived gas it relaxes toward lower values reflecting extended thermalization. Despite its diagnostic power, the OPR has never been measured in the quiescent H_2 gas characteristic of CO-dark environments, as prior facilities lacked the sensitivity to detect the required low-excitation rotational lines. This program will deliver the first high-sensitivity measurements of H_2 rotational emission in a nearby CO-dark cloud, enabling direct determination of the OPR and the first empirical constraint on the chemical age of CO-dark molecular gas.

This Proposal: H_2 Ortho-to-Para Ratio as a Chemical Clock for the Eos Dark Cloud

The newly discovered Eos cloud is the nearest known CO-dark molecular cloud (~94 pc; Burkhart et al. 2025), making it an ideal target for high sensitivity H_2 rotational spectroscopy. Its proximity maximizes surface brightness, reduces confusion with unrelated foreground and background emission, and enables spatial resolution of distinct physical environment within a single CO-dark structure. JWST/NIRSpec uniquely enables this measurement: its combination of sensitivity and spectral resolution is required to detect faint, low-excitation H_2 rotational lines in cold, quiescent gas. **No prior facility could execute this observation.** Spitzer and SOFIA, for example, lacked the sensitivity for low-excitation H_2 lines at the expected column densities ($\sim 10^{20}$ cm^{-2}; Burkhart et al. 2025), and no current facility offers both the required near-infrared throughput and spectral resolution.

<u>We propose deep JWST/NIRSpec spectroscopy targeting the H_2 S(0) and S(1) rotational lines across two regions in Eos</u>: (1) a dense well-shielded region and (2) a diffuse edge region. This two-region strategy is designed to sample the OPR across distinct physical environments and assess whether chemical age varies with shielding depth. The OPR measured in each region will be interpreted using the framework shown in Figure 1, which maps OPR values onto three chemical regimes as a function of time elapsed since the last heating or mixing event: (1) chemically young gas, recently processed or shock-heated; (2) dynamically quiescent but chemically non-equilibrium gas; and (3) fully relaxed, long-lived gas. This framework directly links the observed OPR to the chemical age of Eos, breaking the degeneracy that existing tracers cannot resolve.

This program will produce a direct measurement of the H_2 OPR in a CO-dark molecular cloud, <u>the first such measurement for a CO-dark environment</u>. This measurement will classify Eos into one of the three chemical regimes identified in Figure 1, placing the first empirical constraint on the chemical age and survival timescale of CO-dark molecular gas.

Implications for ISM Evolution and Star Formation

A direct measurement of the H_2 OPR in Eos will provide the <u>first empirical constraint on the chemical age of CO-dark molecular gas</u>, resolving a long-standing observational degeneracy in which existing observational tracers are unable to inform chemical history. This result will critically inform models of molecular cloud assembly and dispersal, establishing whether CO-dark clouds share evolutionary pathways with CO-bright clouds and clarifying their broader role in setting the initial conditions for star formation.

This program is uniquely timely: JWST now enables access to faint, low-excitation H_2 rotational lines in cold quiescent gas for the first time; the recent discovery of Eos provides a clean, nearby laboratory at unprecedented proximity; and theoretical models of intermittent turbulence and time-dependent chemistry now make directly testable predictions against which OPR measurements can be benchmarked. <u>This program directly addresses JWST's star lifecycle and galaxies over time key science themes</u>, and no existing or planned facility can replicate this diagnostic in cold, CO-dark environments. The publicly released JWST/NIRSpec spectra and derived OPR measurements will serve as a community benchmark, anchoring future ISM programs and providing empirical calibration for non-equilibrium chemical network models, star formation initial condition studies, and baryon cycling models in galaxy evolution simulations.

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